2.1 - Matrix Operations

2. $A+2B=\begin{bmatrix}2 & 0 & -1\\ 4 & -3 & 2\end{bmatrix}+2\begin{bmatrix}7 & -5 & 1\\ 1 & -4 & -3\end{bmatrix}=\begin{bmatrix}2+14 & 0-10 & -1+2\\ 4+2 & -3-8 & 2-6\end{bmatrix}=\begin{bmatrix}16 & -10 & 1\\ 6 & -11 & -4\end{bmatrix}$

The expression $3C - E$ is not defined because $3C$ has 2 columns and $-E$ has only 1 column.

$$CB=\begin{bmatrix}1 & 2\\ -2 & 1\end{bmatrix}\begin{bmatrix}7 & -5 & 1\\ 1 & -4 & -3\end{bmatrix}=\begin{bmatrix}1\cdot 7+2\cdot 1 & 1(-5)+2(-4) & 1\cdot 1+2(-3)\\ -2\cdot 7+1\cdot 1 & -2(-5)+1(-4) & -2\cdot 1+1(-3)\end{bmatrix}=\begin{bmatrix}9 & -13 & -5\\ -13 & 6 & -5\end{bmatrix}$$

The product EB is not defined because the number of columns of E does not match the number of rows of B.

4. $A-5I_3=\begin{bmatrix}9 & -1 & 3\\ -8 & 7 & -3\\ -4 & 1 & 8\end{bmatrix}-\begin{bmatrix}5 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}=\begin{bmatrix}4 & -1 & 3\\ -8 & 2 & -3\\ -4 & 1 & 3\end{bmatrix}$

$$(5I_3)A=5(I_3A)=5A=5\begin{bmatrix}9 & -1 & 3\\ -8 & 7 & -3\\ -4 & 1 & 8\end{bmatrix}=\begin{bmatrix}45 & -5 & 15\\ -40 & 35 & -15\\ -20 & 5 & 40\end{bmatrix}, \text{ or}$$

$$(5I_3)A=\begin{bmatrix}5 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}\begin{bmatrix}9 & -1 & 3\\ -8 & 7 & -3\\ -4 & 1 & 8\end{bmatrix}=\begin{bmatrix}5\cdot 9+0+0 & 5(-1)+0+0 & 5\cdot 3+0+0\\ 0+5(-8)+0 & 0+5\cdot 7+0 & 0+5(-3)+0\\ 0+0+5(-4) & 0+0+5\cdot 1 & 0+0+5\cdot 8\end{bmatrix}$$

$$=\begin{bmatrix}45 & -5 & 15\\ -40 & 35 & -15\\ -20 & 5 & 40\end{bmatrix}$$